

L2A™

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(b) Implications of migration in population genetics

Migration refers to the movement from one place to another.

Population genetics popularised by Hardy-Weinberg model studies the genetics over a population and its effects.

Implications

1) Migration results in sudden change of gene frequencies as seen in genetic drift or Sewell Wright effect

2) eg: It may give rise to ~~the~~ a new population with new genetic features.

For eg: Amishi population have Six-fingered Dwarfism common which eventually resulted due to migration

3) Migration may lead to expression of a recessive lethal gene $\Rightarrow$ which may increase homozygosity.

4) It also changes the nature of population and leads to race formation.
eg: Mullatto is result of mating between caucasoid and negroid.

The phenomenon increased after globalization & colonialism giving rise to race crossing.

5) Migration may also increase genetic load of a population due to expression of a deleterious gene.

6) Sometimes, migration also promotes hybridisation. Various factors like natural selection, mutation acts on migration that overall impacts population.